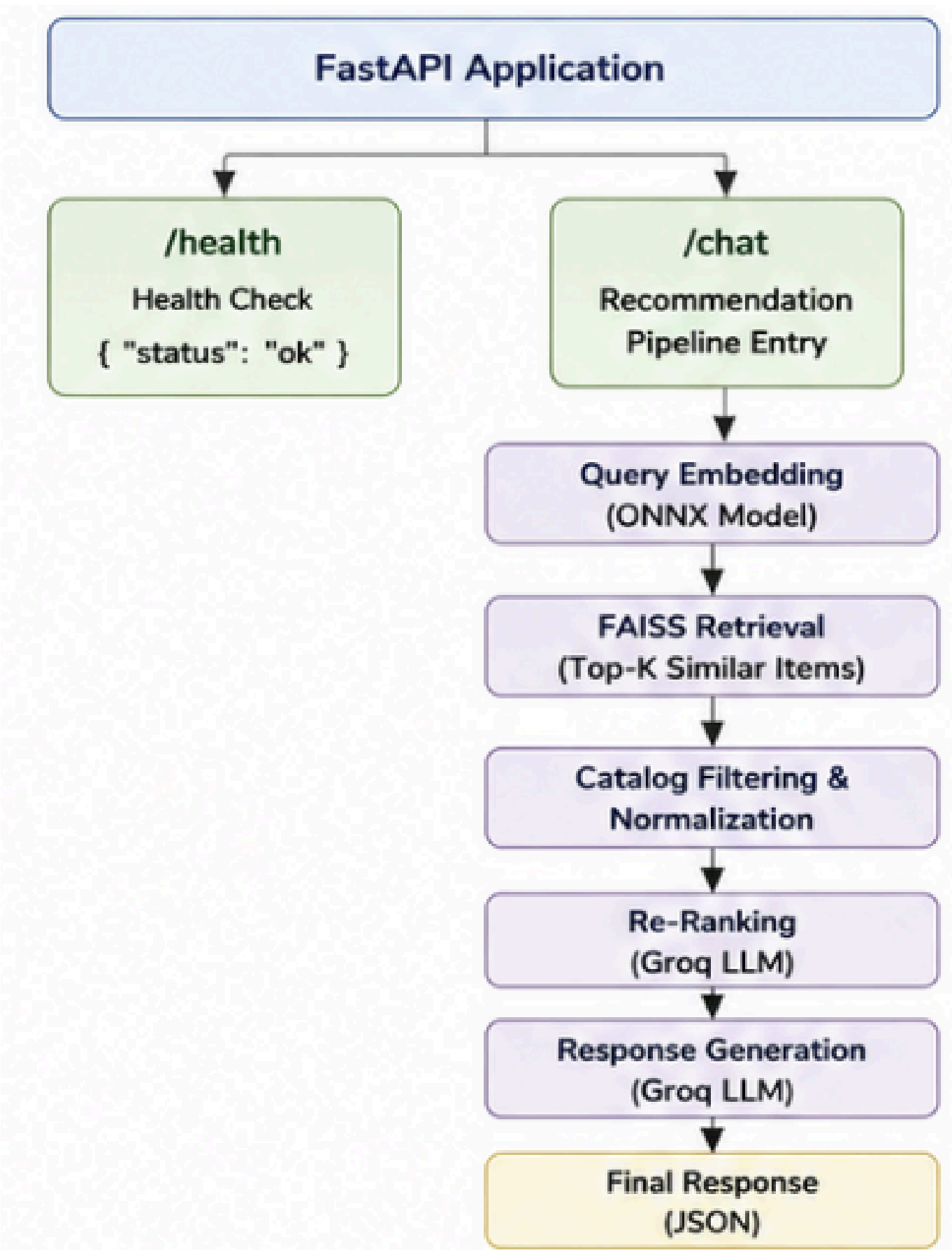


SHL AI Assessment Recommendation API

[Live API](#)

Ridhi Jaggi

System Architecture



System Design Principles

Agility

- Flexible hybrid pipeline (rules + LLM + embeddings)
- Supports both comparison and recommendation flows
- Easily extendable filtering/ranking logic

Fallback & Resilience

- Multi-level fallbacks (LLM → FAISS → catalog)
- Graceful degradation on failures (API, parsing, retrieval)
- Safe SHL-only refusal on invalid queries

Accuracy & Relevance

- Semantic retrieval using FAISS embeddings
- Job-level, skill, and test-type filtering
- Rule-based boosts to correct LLM ranking bias

Safety & Guardrails

- SHL-only scope enforcement
- Injection detection (regex + LLM classifier)
- Hard block for out-of-domain queries

Efficiency & Latency Control

- ONNX embeddings for lightweight inference
- FAISS for fast vector search
- Reduced LLM load via pre-filtering + trimming

Hybrid Intelligence

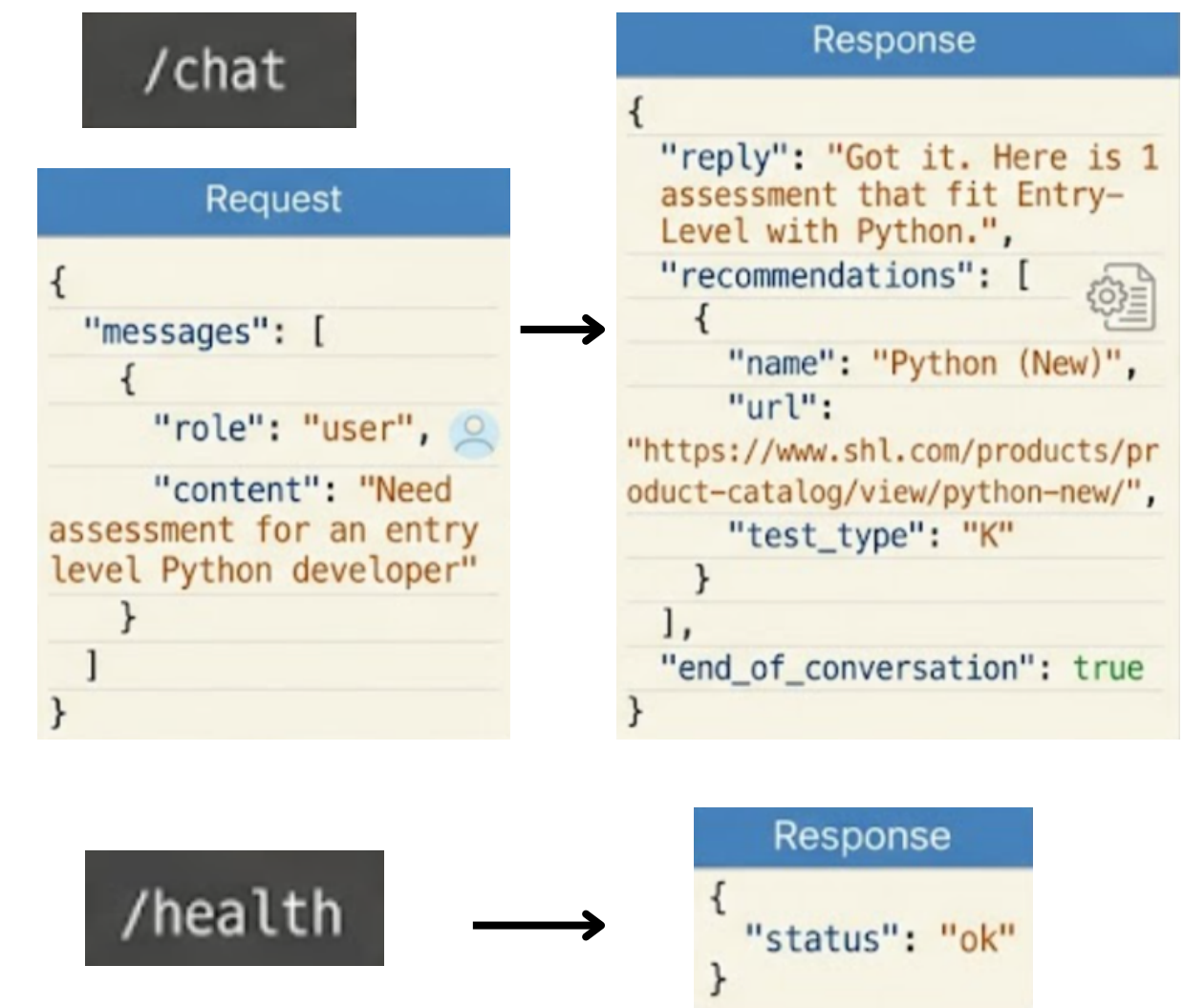
- LLM for reasoning (constraints, ranking, comparison)
- FAISS for semantic retrieval
- Rules for deterministic control and overrides

Fallback & Resilience

- Groq API failure / rate limit** → exponential retry (capped retries) → fallback to safe default response
- Invalid LLM output (malformed JSON)** → JSON cleaning + regex extraction → structured fallback payload
- FAISS retrieval failure / empty results** → fallback to raw catalog or nearest semantic candidates
- Over-filtering (zero results after filters)** → progressively relax constraints → re-run with broader candidate set
- Injection / out-of-scope query** → hard block at safety layer → SHL-only refusal response

Comparison pipeline failure (missing entities) → partial match via semantic search → else ask clarification

API Interface



Technology Used



Prompt Design

1.Issue: when user input was vague, model was still trying to answer instead of stopping
Fix: added rule → if input is vague, respond by asking to clarify role / skill / requirement before doing anything

2.Issue: scope was missing in many cases (what exactly person will do in job was unclear)
Fix: added scope check step before recommendations

3.Issue: it was mixing decision levels (individual contributor vs manager vs stakeholder type roles)
Fix: added rule to clearly separate role type before selecting assessments

3.Issue: tech stack importance was not handled properly (Java / Python / backend skills were not weighted correctly in early versions)
Fix: added strong priority to core tech stack keywords when selecting tests

4.Issue: early-career vs senior mapping was inconsistent in some cases
Fix: added stronger mapping between experience level and assessment difficulty

Final behavior became:

if vague → ask clarification

if clear → check scope + role type + tech stack priority then only recommend assessments

Achieved

- clarification for vague inputs
- scope-first decision flow
- role separation (IC / manager / stakeholder)
- tech stack prioritization
- experience-level mapping

Improvements Needed

- occasional mild hallucination in edge cases
- retrieval can be tighter for catalog-only matching
- some ambiguity in role interpretation still remains
- Evaluation can be improved with stronger semantic/ranking metrics beyond word-overlap.

Evaluation

PS C:\Users\Ridhi\shl> python -u "c:\Users\Ridhi\shl\evaluation.py"

SHL Evaluation Dashboard

Query	GT	Predictions	Recall@K	Latency
We need a solution for senior leadership.	Occupational Personality Questionnaire OPQ32r	Universal Competency Framework Job profiling guide, Universal Competency Framework Profiler Cards (44)	0	1.61
I'm hiring a senior Rust engineer for high-perform	SHL Verify Interactive G+ Programming	Linux Programming (General), Verify - Verbal Ability - Next Generation	1	4.99
We're screening 500 entry-level contact centre age	Entry Level Contact Center Call Simulation (New)	Entry Level Customer Service (General) Solution, Entry Level Customer Serv-Retail & Contact Center	1	34.91
Hiring graduate financial analysts – final-year st	SHL Verify Interactive – Numerical Reasoning	Verify - Numerical Ability, Manufacturing & Industrial - Mechanical Focus 8.0	1	42.36
As part of our restructuring and annual talent aud We're hiring plant operators for a chemical facili We're hiring bilingual healthcare admin staff in S I need to quickly screen admin assistants for Exce Here's the JD for an engineer we need to fill. Can We run a graduate management trainee scheme. We ne	Occupational Personality Questionnaire OPQ32r Manufacturing & industrial - Safety & Dependability 8.0 Dependability and Safety Instrument (DSI) Microsoft Excel 365 (New)		0	33.49
		Dependability and Safety Instrument (DSI)	0	18.89
		Dependability and Safety Instrument (DSI)	1	27.77
		Microsoft 365 (New), Customer Service Phone Solution	1	33.81
		Core Java (Advanced Level) (New), SQL (New)	1	34.59
		Occupational Personality Questionnaire OPQ32r, Verify - Verbal Ability - Next Generation	1	32.32
What is the weather today? Can you check the weather near me and suggest if I			1	0.87
			1	0.41

=== FINAL METRICS ===

"Recall@K": 0.75,

"Groundedness": 1.0,

"total_cases": 12

